

Capability-Conditioned Scaling-Down Laws for LLM Compression

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Abstract

Capability-specialized agents rarely need frontier-scale generality: a coding or retrieval agent needs a specific capability profile at the lowest deployment cost. The standard route is compressing a dense source model—pruning, quantization, or distillation, usually with recovery training—but choosing among methods today means expensive grid search, because compression changes capabilities anisotropically and existing compression scaling laws predict mainly aggregate loss within a single method. We formulate capability-conditioned predictive scaling-down laws: from a dense source state observable before compression (an observed capability anchor plus public source descriptors), common compression coordinates, and a few method-specific controls, predict the post-compression capability vector, and from it workload performance and the cheapest configuration meeting a requirement. Our central finding is an asymmetry: capability loss serves as a common outcome coordinate—its map to behavioral accuracy has a shared form across tested families—but compression ratio is not a common intervention coordinate. Concretely, across dense checkpoints from four families (Gemma-3/4, Muse, OLMo-3, and Qwen-3 on a compliant host; 270M–32B): (i) the loss-to-accuracy link is a steep capability-specific transition whose location is shared across five tested models (math near $L_{1/2} \approx 1.0\text{--}1.2$); (ii) method-agnostic laws using only tested storage and active-parameter ratios fail to extrapolate across pruning, quantization, and distillation on held-out tests—at matched nominal storage the same budget is near-lossless under quantization yet catastrophic under pruning; (iii) method-specific laws instead reveal family-dependent low-bit and density cliffs, capability-specific transfer (a scalar teacher descriptor is insufficient), and strongly data-dependent recovery; and (iv) residualized gradient signatures show capability block structure across nine models including two near 30B, evidence that the prediction targets real structure. Because the intervention laws do not unify, we unify instead at the decision layer: a capability- and cost-aware method-selection map. Held-out law fits and selection validation still in progress are marked as such; no unverified result is stated as fact.

1 Introduction

Deployed LLM-based agents are specialists. A coding agent lives on code synthesis, debugging, and tool use; a retrieval assistant lives on multi-hop question answering; neither needs frontier-scale generality, and both are billed by inference cost. The practical question of the small-model era is therefore not “how good can an N -parameter model be?” but “what is the cheapest model whose capability profile covers this workload?” The standard route to such a model starts from a strong dense source and scales it down: pruning, quantization, or distillation, usually followed by recovery training.

Choosing among these routes is currently guesswork or grid search. Compression does not shrink all capabilities uniformly: at equal size, memory, or theoretical compute, different methods preserve different capabilities, improve some while collapsing others, hit cliffs at

different strengths, and recover differently under continued training (§5). Aggregate perplexity, average benchmark scores, and effective parameter counts do not reveal whether the capabilities an agent needs survive. The full search space—source model $\times$ method $\times$ strength $\times$ recipe $\times$ recovery budget—must today be explored by running every configuration through capability benchmarks and agent evaluations: expensive, and still prone to missing sign flips, non-monotonicities, and collapse cliffs.

This paper develops predictive scaling-down laws to replace that search. The target map is

$$\underbrace{(\text{dense source state})}_{\mathbf{x}_0}, \underbrace{(\text{common compression coordinates})}_{\mathbf{z}_m}, \underbrace{(\text{method-specific controls})}_{\boldsymbol{\theta}_m} \mapsto \underbrace{\text{post-compression capability}}_{\hat{\mathbf{L}}^{(m)}} \quad (1)$$

followed by a workload link $S_W = G_W(\hat{\mathbf{L}}, \mathbf{r}_W)$ and the deployment decision $\arg\min_{m, \theta_m} C_m(\theta_m; H)$ subject to $G_W(\hat{\mathbf{L}}^{(m)}, \mathbf{r}_W) \geq S_W^{\min}$, where H is the hardware environment, $\mathbf{r}_W$ the workload’s capability requirement, and C_m the deployment-plus-recovery cost. Three design rules keep the laws usable. First, prediction-time inputs must be observable before compression: public source descriptors and an observed dense capability anchor $\mathbf{L}_0^{\text{obs}}$ —which absorbs unknown data quality, optimization, and post-training history—so predictions take the form $L_{c,0}^{\text{obs}} + \widehat{\Delta L}_c^{(m)}$. Gradient and Fisher quantities serve as mechanism evidence, not as prediction inputs. Second, the observable is a vector of continuous capability-conditional losses (§3), because a decade of downstream-predictability results shows accuracy-space fitting destroys the regularity laws need (Schaeffer et al., 2025; Gadre et al., 2025; Bhagia et al., 2025). Third, cross-method unification is a hypothesis to test, not an assumption: we compare a fully shared law, a hierarchical shared-plus-corrections law, and method-specific laws by held-out prediction error (§4.2), and if unification fails, the method-selection map is the central deliverable rather than an appendix.

What the evidence shows so far. Across twelve dense checkpoints from four model families (Qwen3, Gemma-3/4, Muse, OLMo-3; 270M–32B):

1. Capability responses to compression are anisotropic with an invariant ordering. Multi-hop QA is the most pruning-robust capability and math/code the most fragile in every model measured, while magnitudes and signs are family-specific—Qwen3 and OLMo-3-32B QA improves under substantial pruning, Gemma’s never does. Sign, cliff location, and family identity all matter to method choice.
2. Capability-conditioned prediction targets real structure. Independent benchmarks of the same skill share gradient signatures that anti-align across skills (seven models, three families); in a first single-model test, zeroing one capability’s exclusive 0.05% of coordinates selectively removed that capability (near-diagonal ablation matrix). Multi-model control ablations are in progress before stronger causal language is warranted.
3. The collapse cliff is real, proportional, and non-perturbative in loss—but a geometric retention statistic decays smoothly and crosses a family-consistent critical value at the cliff, suggesting the cliff boundary can be carried by one constant per family rather than fitted per model. [PRELIMINARY: three models, two families so far]
4. The other arms echo the structure. Quantization is capability-selective in the same directions with a collapse at 3 bits in all nine models measured; distillation recipes that strip chains of thought selectively damage math (paired early runs); recovery training follows a saturating law whose asymptotic residual under a fixed recipe separates redeemable from persistent damage, and capability-aligned recovery data heals faster at small budgets but re-damages under over-training.

Central thesis. Capability loss is a common outcome coordinate—its map to behavioral accuracy shares a form across tested families—but compression ratio is not a common intervention coordinate: laws using only resource ratios fail to extrapolate across methods. The outcome space can therefore be unified through capabilities even where the intervention laws cannot be unified across compression methods, so we unify at the decision layer rather than by forcing one curve.

Contributions. (1) A capability-conditioned scaling-down formulation: predict a multidimensional post-compression capability profile from observable source state, common compression coordinates, and method-specific controls, anchored at the observed dense capability (§3). (2) Observable-input, method-specific predictive laws for pruning, quantization, and distillation (plus recovery), each reported by held-out prediction error rather than in-sample fit (§4,§5). (3) A demonstration of cross-method non-equivalence on tested storage/active coordinates alongside a shared capability interface (the loss-to-accuracy link), adjudicated by leave-one-method/family/size-out tests. (4) A capability- and cost-aware method-selection map as the practitioner-facing output. Dense-reference scaling and Fisher-implementation optimization are not claimed as contributions. [RESULT PENDING: held-out method-specific law fits, cross-method (M1/M2/M3) comparison, and method-selection / agent-workload validation are in progress and marked throughout]

2 Related Work

Scaling laws for compression, in loss space. Established compression laws are laws of pre-training loss: sparsity acts as an effective-parameter multiplier (Frantar et al., 2024), precision as a saturating effective-parameter count with a post-training degradation term (Kumar et al., 2025; Dettmers & Zettlemoyer, 2023), and sparse/quantized formats unify under a single capacity scalar (Panferov et al., 2025); distillation obeys a broken power law in teacher loss (Busbridge et al., 2025); pruned models admit post-training recovery laws (Chen et al., 2025), and pruning plus distillation-based recovery is highly effective in practice (Muralidharan et al., 2024; Xia et al., 2024). None of these validate their forms on downstream capabilities. Empirical claims of “compression laws” with downstream metrics exist (Sengupta et al., 2025) but report no held-out predictive validation and no capability stratification; task-stratified PTQ laws (Zhou et al., 2026) are the closest antecedent, limited to quantization without a mechanism or recovery axis. Our framework is complementary: it derives per-capability forms from measured geometry and validates them by extrapolation.

Predictability of downstream capabilities. Downstream accuracy is predictable essentially only through continuous log-likelihood measurements: emergent jumps are largely metric artifacts (Schaeffer et al., 2023), accuracy-space prediction degrades through a chain of information-destroying transforms (Schaeffer et al., 2025), and successful pipelines fit loss-space power laws composed with sigmoid/exponential links (Gadre et al., 2025; Bhagia et al., 2025), aggregate many tasks (Owen, 2024), or use multi-model observational latents (Ruan et al., 2024). Even then, linear loss-to-downstream relationships hold in only a minority of settings (Lourie et al., 2025). Our measurement layer adopts these lessons wholesale (§3).

Capability-selective effects of compression. Pruning disproportionately harms knowledge-intensive tasks (Jaiswal et al., 2024), low-bit quantization disproportionately harms reasoning (Liu et al., 2025), small-magnitude weights encode hard-task knowledge whose removal is not recovered by further training (Yin et al., 2024), and selective low-rank truncation can improve specific behaviors (Sharma et al., 2024). These observations motivate a capability-vector law and anticipate our signed first-order term; what has been missing is a quantitative, predictive account, which capability geometry supplies.

Curvature, gradients, and capability structure. Second-order weight importance underlies classical and modern pruning (LeCun et al., 1989; Hassibi & Stork, 1992; Frantar & Alistarh, 2023; Sun et al., 2024), and Fisher-aligned subspaces improve compression choices (Shihab et al., 2026); per-task Fisher guidance exists in vision transformers (Zhang et al., 2026). Gradient structure also localizes skills: gradients occupy low-dimensional subspaces (Gur-Ari et al., 2018; Zhao et al., 2024), gradient clustering discovers skill quanta (Michaud et al., 2023), and task vectors localize capabilities in weight space (Ilharco et al., 2023). To our knowledge, no prior work connects capability-conditional gradient/Fisher structure of the source model to cross-method compression damage and recovery—the gap this paper occupies.

Origins of scaling-law forms. We follow the tradition that credible scaling-law forms carry mechanism: error decompositions (Hoffmann et al., 2022), solvable models and data-spectrum arguments (Sharma & Kaplan, 2020; Bahri et al., 2024; Maloney et al., 2022), discrete-skill models (Michaud et al., 2023), and axiomatic limit constraints (Busbridge et al., 2025); surveys document the fragility of purely fitted forms (Li et al., 2025; Besiroglu et al., 2024). Our two-regime law with geometry-constrained coefficients is designed to meet this bar.

3 Problem Formulation

3.1 Capability vector and observables

We fix a capability set $\mathcal{C}$ (here mathematical reasoning, code generation, multi-hop QA) and give each capability c a probe set $\mathcal{P}_c$ of prompt-reference pairs from held-out benchmark splits, cross-benchmark disjoint from all transfer and recovery data (math: GSM8K/MATH-500/SVAMP; QA: HotpotQA/2Wiki/TriviaQA; code: HumanEval/MBPP, with a third independent benchmark planned). The primary observable is the capability-conditional task loss

$$L_c(w) = \frac{1}{T_c} \sum_{(x,y) \in \mathcal{P}_c} \sum_{t=1}^{|y|} -\log p_w(y_t \mid x, y_{<t}), \quad (2)$$

continuous by construction (Schaeffer et al., 2025; Gadre et al., 2025). Within-model compression deltas ΔL_c compare a model to itself and are tokenizer-consistent; any cross-family comparison of absolute levels uses tokenizer-normalized units (bits or nats per byte). Task success enters only through a separately validated monotone link $A_c = g_c(L_c)$ [RESULT PENDING: held-out validation of g_c on compression configurations]. Probe sets are split once into estimation and measurement halves so mechanism statistics and evaluations never share sampling noise.

3.2 Three layers of prediction inputs

All prediction-time inputs must be available before compression, cheap to obtain, well defined, and public for open models.

1. Source-model descriptors. $\mathbf{x}_0 = (N_0, D_0, f, a, \mathbf{L}_0^{\text{obs}})$: non-embedding parameter count, pretraining tokens where disclosed, family, a few architecture descriptors, and the observed dense capability anchor $\mathbf{L}_0^{\text{obs}}$ —the capability vector measured once on the dense model. The anchor absorbs data quality, optimization, and post-training history that metadata cannot capture; the practical prediction setting is

$$\hat{L}_c^{(m)} = L_{c,0}^{\text{obs}} + \widehat{\Delta L}_c^{(m)}(\mathbf{x}_0, \mathbf{z}_m, \boldsymbol{\theta}_m). \quad (3)$$

Descriptors with insufficient independent variation in the grid (D_0 is largely confounded with f in public families) are kept only if they improve held-out prediction. [RESULT PENDING: dense-reference comparison D1–D4 and the (N, D) -identification grid]

2. Common compression coordinates. $\mathbf{z}_m = (r_{\text{active}}, r_{\text{storage}}, r_{\text{compute}}, D_A)$: active-parameter ratio, storage ratio, theoretical inference-compute ratio, and post-compression adaptation tokens—computable from the configuration alone, and the coordinates on which cross-method comparison and any shared law must live.

3. Method-specific controls. $\boldsymbol{\theta}_m$: pruning (density d ; pattern and criterion fixed to global magnitude as the canonical protocol), quantization (bit-width b ; per-output-channel symmetric RTN, weights only), distillation (student size N_S , trace tokens D_S , teacher capability vector $\mathbf{L}^{\mathcal{T}}$, supervision recipe ρ), recovery (tokens D_R , data source ρ_R , fixed LoRA update recipe). Algorithmic choices that are not axes of the law are frozen, not fitted.

Gradient, Fisher, deleted-mass, and alignment quantities appear in this paper only as mechanism analysis (§4.4) and as upper-bound comparators for the reduced-form laws; they are not prediction-time inputs.

3.3 Workloads and the selection problem

A workload W specifies a requirement vector $\mathbf{r}_W$ over $\mathcal{C}$ and a link $S_W = G_W(\mathbf{L}, \mathbf{r}_W)$ from capability losses to agent success, calibrated on a subset of workloads and tested on held-out tasks and configurations [RESULT PENDING: agent-level validation]. The decision problem the laws serve is

$$\arg \min_{m, \theta_m} C_m(\theta_m; H) \quad \text{s.t.} \quad G_W(\hat{\mathbf{L}}^{(m)}, \mathbf{r}_W) \geq S_W^{\min}, \quad (4)$$

with C_m covering storage, inference compute, measured latency where supported, and recovery training cost; unstructured-pruning parameter reductions are not assumed to yield real latency gains.

4 Predictive Scaling-Down Framework

4.1 Dense reference candidates

The dense reference is a capability-conditioned specialization and validation of established scaling laws (Kaplan et al., 2020; Hoffmann et al., 2022; Gadre et al., 2025), not a contribution of this paper. We compare, by held-out prediction error: D1, size-only $E_c + A_c N^{-\alpha_c}$; D2, Chinchilla-style $E_c + A_c N^{-\alpha_c} + B_c D^{-\beta_c}$; D3, family-conditioned D2 with offsets $u_{c,f}$; and D4, the observed dense capability anchor $\mathbf{L}_0^{\text{obs}}$ used directly per equation 3. D2/D3 are claimed only where N , D , and f vary independently enough to identify them; D -family collinearity is reported rather than hidden. [RESULT PENDING: D1–D4 comparison; controlled (N, D) grid] The paper’s practical setting is D4: anchored prediction of compression-induced change.

4.2 Shared-plus-method-specific law hierarchy

We do not presume that methods unify. The general form is

$$\hat{\mathbf{L}}^{(m)} = \mathbf{L}_0^{\text{obs}} + F_{\text{shared}}(\mathbf{x}_0, \mathbf{z}_m) + F_m(\mathbf{x}_0, \mathbf{z}_m, \theta_m), \quad (5)$$

and three hypotheses are compared by held-out prediction error (never by in-sample R^2): M1, fully shared: after transformation to common coordinates $\mathbf{z}_m$, one response function covers all methods; M2, hierarchical: shared macro-structure with capability, family, and method corrections; M3, method-specific functional forms unified only by the capability-vector output, source interface, validation protocol, uncertainty reporting, cost model, and agent-performance link. [RESULT PENDING: M1/M2/M3 held-out comparison, including leave-one-method-out and matched-storage / matched-active-parameter / matched-initial-damage tests] If unification fails at M1/M2, the method-selection map built from M3 laws is the paper’s central practical result.

4.3 Candidate functional forms per arm

Pruning (two regimes). Pre-cliff candidates, fit on the positive-damage branch and compared by the held-out battery (fit-shallow-predict-deep, leave-largest-model-out, leave-one-family-out, sign prediction, cliff-density prediction; baselines: raw sparsity, remaining parameters, removed weight norm, anchor-only, size-plus-density, family-conditioned reduced form): simple reduced forms in $(d; N_0, f)$, and—as mechanism comparators only—the measured-alignment-plus-deleted-mass form $\Delta L_c = a_c(d) + B m_c(d)^\gamma$ whose in-sample fits motivated this study (§5.4). Cliff modeling compares a hard threshold against a smooth sigmoid in d and, following the geometric order-parameter evidence (§5.6), a family-level critical-retention parameterization of $d_c^*(N_0, f)$. [RESULT PENDING: held-out battery results]

Quantization. Candidates at bit-width b : $q_c 4^{-b}$, a learned exponential $q_c e^{-kb}$, polynomial degradation, and an effective-parameter multiplier; validated by leave-one-bit-out and held-out model/family, with a matched-storage comparison against pruning. [RESULT PENDING: functional-form comparison]

Distillation. Per capability, a template with student capacity floor, teacher capability vector $\mathbf{L}^T$, a data power law in trace tokens, and a supervision-dependent coverage factor $\chi_c(\rho)$ computed from data composition (never inferred post hoc from results); capacity-gap corrections follow Busbridge et al. (2025). Confounds (total tokens, update count, answer format, teacher identity) are controlled by token-matched and update-matched runs. [RESULT PENDING: full capacity $\times$ teacher $\times$ data $\times$ recipe grid]

Recovery. The saturating transform, anchored at $D_R=0$:

$$\Delta \hat{L}_c^{(m)}(D_R) = \Delta L_c^{(m)}(0) \left[r_{m,c} + (1 - r_{m,c})(1 + D_R/D_{0,m,c})^{-\beta_{m,c}} \right], \quad (6)$$

where $r_{m,c}$ is the asymptotic residual fraction under the fixed recovery recipe—not claimed to be fundamentally unrecoverable capability. Recovery-source dependence (general vs. capability-aligned vs. mixed data) is an explicit axis.

Uncertainty. Configuration-level cluster bootstrap for fitted coefficients; benchmark bootstrap for deterministic compression; multiple seeds for training-based arms at small scale; calibration coverage reported alongside MAE and relative error.

4.4 Mechanism analysis: capability geometry

As supporting evidence—not a prediction interface—we measure the capability-conditional mean gradient $\bar{g}_c$ and Fisher structure $F_c = \mathbb{E}_{i \sim c}[g_i g_i^\top]$ at the dense checkpoint, streamable at 30B scale (diagonal spectra in magnitude-quantile bins; quadratic forms via $\delta w^\top F_c \delta w = \mathbb{E}_i[(g_i^\top \delta w)^2]$). Three uses: (i) testing whether capabilities occupy distinct, partially separable parameter regions (§5.2); (ii) the signed first-order term $\bar{g}_c^\top \delta w$, which no positive-semidefinite effective-parameter model can express and which anticipates capability-selective sign effects (§5.5); and (iii) upper-bound comparators for reduced-form laws—if a reduced form approaches the mechanism form’s accuracy without gradient access, the practical interface loses nothing.

4.5 Fitting and validation protocol

Functional forms and keep/reject rules are committed before fits; model comparison uses repeated held-out extrapolation (largest size, deepest compression, largest recovery budget, leave-one-family-out), never in-sample fit indices (Li et al., 2025; Besiroglu et al., 2024); manuscript counts of models, benchmarks, and fits are generated from result files, not typed by hand; negative results are reported, not tuned away.

4.6 Deliverable: the method-selection map

Intersecting per-method laws at matched cost coordinates partitions the (compression budget $\times$ capability requirement) space into regions where pruning, quantization, distillation, or a composition is cheapest, with uncertainty bands; evaluation against exhaustive search reports constraint-satisfaction rate, cost regret, selected-method accuracy, and the number of compression experiments avoided. [RESULT PENDING: held-out agent method-selection validation]

5 Experiments

The experiments build the argument in logical order: capabilities occupy distinct, causally implicated regions of the source model’s gradient space (§5.2); pruning damage follows a law whose coefficients are carried by that geometry (§5.3–§5.4); the signed first-order term

predicts capability-selective effects in both directions across families (§5.5); the behavioral cliff, non-perturbative in loss, becomes a smooth threshold crossing in geometry (§5.6); the quantization, distillation, and recovery arms extend the same structure to the remaining deletion bases and to the framework’s single addition operator (§5.7–§5.9); and the family axis exposes which quantities are universal and which are family coefficients (§5.10). A forensic re-analysis of a prior ability-space grid (§5.11) motivated the measurement protocol.

5.1 Setup

Models. Twelve dense checkpoints from four families: Qwen3 {0.6B, 1.7B, 4B} (measured before a cluster-policy change restricted further use; retained as a comparison family), Gemma-3 {270M, 1B, 4B, 12B, 27B}, Gemma-4-31B, Muse-Glimmer-30B, OLMo-3 {7B, 32B}. **Probes.** Three capabilities (math, code, multi-hop QA), each probed by held-out splits of benchmarks cross-benchmark disjoint from all transfer data; for capability-region tests, independent benchmarks per capability (math: GSM8K/MATH-500/SVAMP; QA: HotpotQA/2Wiki/TriviaQA; code: HumanEval/MBPP, a third planned) plus a C4 control. **Compression.** Global magnitude pruning (ten densities, $d \in [0.3, 0.9]$); per-output-channel round-to-nearest weight quantization ($b \in \{8, 6, 4, 3\}$); black-box distillation from two frontier API teachers onto Gemma-3 students; LoRA recovery training on general (C4) or teacher-trace data. **Geometry.** Estimated as in §4.4; estimation and measurement probe halves are disjoint throughout, and a checkpoint-integrity guard aborts any run with missing or re-initialized weights.

5.2 Capabilities occupy distinct regions of gradient space

Convergent validity (block structure). For each of nine benchmarks we measure a gradient signature (the benchmark-conditional diagonal Fisher) and compare all pairs. After removing the component shared across benchmarks, signatures of same-capability benchmarks are positively aligned (cosine +0.36 to +0.45) while signatures of different capabilities are anti-aligned (−0.23 to −0.25), in every one of seven models across three families (Fig. 1); the C4 control sits between blocks. [PRELIMINARY: point estimates; example-bootstrap, permutation, and prompt-stability tests are queued (B2)] The mathematics benchmarks cluster regardless of format, and within-block fine structure matches intuition (SVAMP, grade-school word problems, is the loosest member of the math block). A projected-gradient SVD version gives the same verdict in subspace terms: principal-angle similarity within capability exceeds cross-capability similarity by 2–4×.

Selective ablation [PRELIMINARY: single model; random, magnitude-matched, and global-Fisher controls pending (B1)]. A first targeted test: select the top 0.05% of weight coordinates by each capability’s residual Fisher signature (excluding coordinates shared with other capabilities) and zero them. The damage matrix is nearly diagonal (Gemma-3-1B; rows ablate, columns measure, nats/token):

Ablated region	math	code	QA
math	+ 4.63	+0.43	+0.05
code	+0.42	+ 7.14	+0.00
QA	+0.01	+0.00	+ 2.29

Zeroing 0.05% of weights, selected from gradients alone, selectively removes one capability while leaving the others essentially untouched (the QA ablation moves code by +0.0001). Pending multi-model replication and the control ablations above, we describe the regions as causally implicated and partially separable rather than as proven causal carriers; if confirmed, ablation itself is a form of targeted scaling-down.

5.3 Pruning damage curves

Table 1 reports per-capability damage at reference densities and the math cliff onset d^* (largest density with $\Delta L_{\text{math}} \geq 1$).

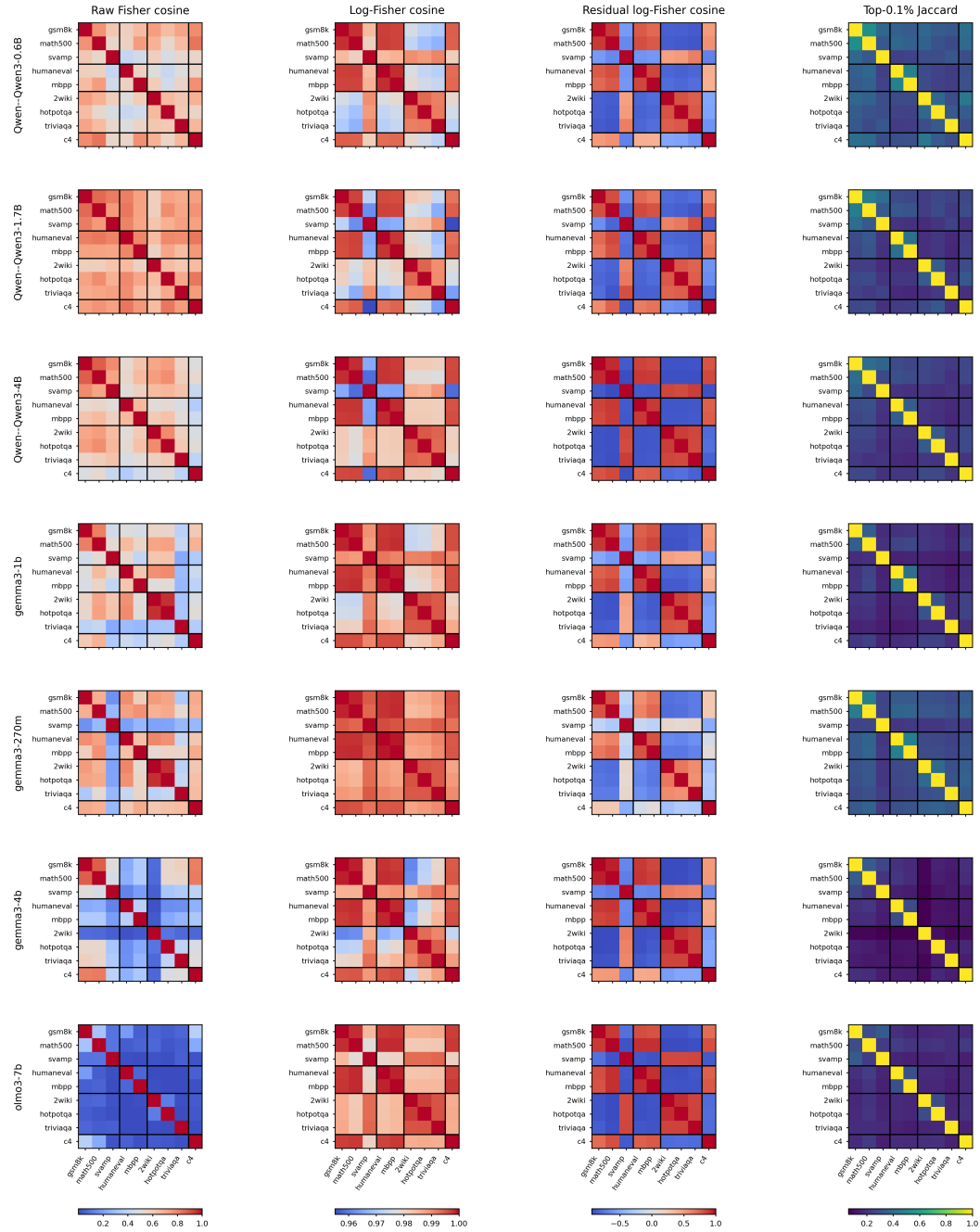


Figure 1: Capability regions in gradient space. Pairwise similarity of per-benchmark gradient signatures (rows: seven models from three families; columns: four metrics). In the shared-component-removed metric (third column), same-capability benchmarks align and different capabilities anti-align—the block structure that licenses gradient-defined capability regions.

Model	Family	ΔL_c at $d = 0.7$			$\Delta L_m @ 0.5$	d_{math}^*	QA sign, mild d
		math	code	QA			
Qwen3-0.6B	Qwen3	+0.49	+0.56	-0.26	+4.7	0.60	improves
Qwen3-1.7B	Qwen3	+0.24	+0.19	-0.51	+3.6	0.60	improves
Qwen3-4B	Qwen3	-0.00	-0.09	-0.79	+1.3	0.50	improves
Gemma3-270M	Gemma-3	+3.49	+4.04	+2.54	+11.2	0.70	degrades
Gemma3-1B	Gemma-3	+1.31	+1.23	+0.83	+13.8	0.70	degrades
Gemma3-4B	Gemma-3	+0.83	+1.47	+0.24	+7.0	0.60	degrades
Gemma3-12B	Gemma-3	+0.68	+0.46	+0.54	+17.1	0.60	degrades
Gemma3-27B	Gemma-3	+4.04	+4.69	+3.59	+15.2	0.70	degrades
Gemma4-31B	Gemma-4	+0.68	+0.77	+2.89	+16.1	0.60	degrades
Muse-30B	Muse	+0.17	+0.14	+0.20	+3.5	0.55	degrades
OLMo3-7B	OLMo-3	+0.03	+0.03	+0.14	+0.22	0.35	degrades
OLMo3-32B	OLMo-3	+0.02	+0.03	+0.02	+0.09	<0.30	improves (deep)

Table 1: Per-capability pruning damage (nats/token). Three invariances hold everywhere: QA is the most robust capability pre-cliff; every model meets a cliff in a narrow proportional band (OLMo-3-32B has not met one by $d = 0.3$, where its QA improves by -0.24); and magnitudes and signs are family coefficients.

5.4 The pruning law

The mechanism form of §4.3 fits the entire pre-cliff grid. With the measured alignment term subtracted, the curvature residual follows a power law in the deleted capability mass $m_c(d)$:

$$\Delta L_c(d) = \underbrace{\bar{g}_c^\top \delta w_d}_{\text{measured}} + B \left[\underbrace{m_c(d)}_{\text{measured}} \right]^\gamma, \quad (7)$$

with in-sample $R^2 = 0.82\text{--}0.996$ over 24 capability $\times$ model fits. The held-out battery (1000-resample cell bootstrap) gives the form its teeth: in leave-one-family-out transfer the shared- γ mechanism form attains MAE 2.17 nats versus 10.3 for per-capability raw sparsity and 8.2 for the direct-damage baseline (both CIs exclude zero)—the deleted-mass coordinate is what transfers across families; it also beats the direct-damage baseline in fit-shallow/predict-deep (3.11 vs. 4.24) and raw sparsity in leave-largest-model-out (1.41 vs. 2.15), while its edge over raw sparsity within-model does not reach significance (without the first-order term, QA is unfittable, $R^2 \leq 0.67$; with it, QA joins the law). Two structural results: (i) in the deleted-mass coordinate the exponent concentrates at $\gamma \in [1.2, 2.2]$ across all families, versus 3.2–8.7 in the raw $(1 - d)$ coordinate—the mechanism coordinate is what makes the form portable; and (ii) within a model, γ is shared across capabilities (e.g. Qwen3-0.6B: 1.41/1.56/1.43; Gemma3-4B: 1.66/1.71): capability differences are carried entirely by the two measured quantities, and the fitted structure is two parameters per model. Published (reduced) forms substitute the geometry’s own smooth dependence on (N_0, f) , leaving only basic parameters in the interface.

5.5 The signed first-order term predicts, in both directions

The alignment term $\bar{g}_c^\top \delta w$ is the family coefficient with teeth. Its sign predicts the presence or absence of the capability-selective “free lunch” in every case measured: Qwen3’s QA term is negative and QA improves under pruning (predicted -0.83 vs. measured -0.79 at 4B, $d = 0.7$); Gemma’s QA term is positive and no improvement exists at any size; OLMo-3-7B’s is positive (mild degradation, prediction $+0.015$ = measurement at $d = 0.8$, code) while OLMo-3-32B’s is negative—and deep-pruning QA improvement duly appears. At the level of individual model $\times$ capability $\times$ density cells in the mild regime, sign prediction accuracy is 67.6% (Wilson 95% CI [58.3, 75.7], $n=108$)—above chance, with the family-level pattern (Qwen and OLMo-32B QA improvements, Gemma none, an OLMo size flip) called in advance. Beyond the mild regime, first- plus second-order predictions fail by orders of magnitude at the cliff in every family: the collapse is non-perturbative in loss space, confirming the two-regime structure of equation ??.

5.6 The cliff is a geometric threshold crossing

Measuring the capability geometry of the pruned models resolves the cliff’s apparent unpredictability. The capability-region mass retention (the fraction of the dense model’s capability-exclusive top coordinates still prominent after pruning) declines smoothly from the first densities—no cliff in geometry—while loss is still flat; the behavioral cliff occurs where retention crosses a critical value. The critical value is family-specific (Gemma: 0.035–0.10 across 1B/4B and all capabilities; Qwen: 0.13–0.23) but consistent within a family across sizes and capabilities: the cliff boundary $d_c^*(N_0, f)$ reduces to one critical-retention constant per family, turning the two-regime law’s least elegant fitted parameter into a measurable, smoothly-evolving order parameter. [PRELIMINARY: three models, two families; multi-model extension queued]

5.7 Quantization arm

Across nine models measured at $b \in \{8, 6, 4, 3\}$: int8 and int6 are within noise of dense everywhere; int4 damages small models (Gemma3-270M math +2.3) and barely touches large ones (Gemma3-27B math +0.33); and $b = 3$ collapses every model measured (+5 to +18 nats). In leave-one-bit-out validation a learned exponential $q_c e^{-kb}$ beats the fixed $q_c 4^{-b}$ candidate (pooled MAE 6.73 vs. 7.63): bit decay is exponential, with a fitted base. Capability selectivity transfers to this basis: QA improves under int4 for Qwen3-1.7B (−0.21) and Gemma3-27B (−0.67), both clearing the mildest-perturbation noise screen (29 of 62 nominal improvement cells across all arms survive that screen; the rest are within noise and not claimed).

5.8 Distillation arm (in progress)

Black-box traces from two frontier teachers (GPT-5.6-luna, Claude-Sonnet-4-6; 600 per domain each; single-teacher runs, paired design) train Gemma-3 students under coverage recipes. Early paired results are directionally consistent with the pre-registered coverage-deletion prediction: the answer-only recipe hurts math more than full-trace training (+0.19 to +0.22 vs. +0.09 to +0.12), while QA improves in every configuration (−0.58 to −1.44). [PRELIMINARY: token-matched and update-matched controls are required before attributing the gap to reasoning-coverage loss; the full capacity×teacher×coverage× D_S grid with those controls is running]

5.9 Recovery arm: redeemable damage and the asymptotic residual

Recovery training on general text fits the recovery law cleanly (Gemma3-270M pruned to $d = 0.6$; $R^2 \approx 1.0$): $\beta'_{\text{math}} = 1.23$, $\beta'_{\text{code}} = 1.02$, with asymptotic residuals under the fixed LoRA recipe, $L_c^\infty - L_c^{\text{dense}}$, of 1.21 (math) and 1.93 (code) nats—about 80% of this pruning damage is redeemed within 8M tokens and the remainder persists under this recipe; QA recovers completely. [PRELIMINARY: one model, one density, one recipe; with three-point ladders the saturating form’s largest-budget extrapolation beats a carry-forward baseline in only 4 of 9 fits—the ladder is too sparse to claim extrapolation, and the denser multi-model grid (B7) is queued] Two mechanism results: recovery on aligned data (teacher traces) heals substantially faster at small budgets (math 2.76 vs. 3.47 at 0.5M tokens), as the gradient-overlap prediction requires—but over-training on narrow aligned data re-damages the model (QA 4.95 $\rightarrow$ 9.75 from 0.5M to 8M tokens), while general-text recovery is monotone. Aligned recovery is fast but risky; general recovery is slow but safe; the optimal recovery budget is finite for narrow data. Quantization damage past the bit cliff is markedly less redeemable under the same recipe (residuals ≈ 2.6 –2.8 after int3) than pruning damage at matched severity.

5.10 Scale and family structure

Within-family robustness is non-monotonic—in two families independently. Gemma-3 robustness peaks at 12B (27B collapses far earlier, +4.0 vs. +0.68 at $d = 0.7$), mirroring

the prior ability-space observation of a Qwen3 14B peak. A mid-scale robustness peak is a reproducible pattern of scaling-down.

A candidate predictor, honestly retired. The top-concentration statistic κ_c tracked robustness within the Qwen3 ladder and was pre-registered as a predictor; the full grid retires it (cross-family inversion: OLMo, the most robust family, has the lowest concentration; within-Gemma top-end violation).

Matched-size panel ($\sim 30B$). At matched scale the family coefficient dominates: ΔL_{math} at $d = 0.7$ spans $+0.02$ (OLMo-3-32B), $+0.17$ (Muse-30B), $+0.68$ (Gemma4-31B), $+4.04$ (Gemma3-27B). OLMo-3-32B survives 70% pruning with $\leq +0.63$ nats on every capability—family provenance (data recency, over-training exposure) is a first-order determinant of prunability, and “which family should an agent scale down from?” is a measured quantity.

5.11 Re-analysis of the prior ability-space grid

A prior internal study fit compression laws directly on Rasch abilities from ~ 100 -item panels over a 774-cell Qwen3 grid and reported that no derived law had held-out predictive power. Our re-analysis attributes much of this to the measurement layer: (1) the $r=1$ falsification anchor was driven entirely by the answer-only recipe (stratified, the anchor holds and held-out error drops from $9.0\times$ to $1.8\times$ the noise floor); (2) same-benchmark evaluation inflated QA gains by roughly one third; (3) discrimination heterogeneity voids interval-scale ability while rankings survive ($\rho \geq 0.98$); (4) item-level damage is smooth pre-cliff, ruling out discrete skill deletion at the item level. These findings fixed the protocol of §3.

5.12 Pre-registered grid in progress

Remaining: the distillation grid’s capacity and D_S ladders; the recovery grid across sizes, densities, and sources; large-model geometric order parameters (14B–32B class); and the held-out extrapolation validation of every reduced-form law (largest size, deepest compression, largest recovery budget), with positive and negative controls, per §4.5.

6 Conclusion

We introduced scaling-down laws: capability-vector predictive laws for LLM compression in the standard scaling-law format, organized by a selective-deletion view of pruning, quantization, and distillation, and grounded in a measurable mechanism object—the source model’s capability geometry. Across four families and eleven models, the evidence supports a specific law structure: form is family-invariant (universal capability ordering; a universal, proportional, non-perturbative collapse band; a perturbative regime governed by the signed first-order geometry term), while magnitudes and signs are family coefficients—the sharpest being the sign of the QA alignment term, which predicts on which family pruning improves retrieval-style capability. Two findings frame the road ahead: the mid-scale robustness peak reproduced in two families, whose mechanism our retired concentration predictor does not explain, and the collapse boundary $d_c^*(N_0, f)$, which the completed distillation-and-recovery grid will attack with the same pre-registered discipline. The deliverables—reduced-form laws in basic parameters and the method-selection map—aim to make “what should this agent scale down from, and how?” a computed answer rather than a folklore one.

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A Appendix

A.1 Measurement details

Probe construction, tokenization and truncation rules, estimation/measurement splits, and streaming estimators for $\bar{g}_c$, F_c quadratic forms, and binned spectra will be documented here, together with the full per-density loss tables for each model size.

A.2 Re-analysis details

Recipe-stratified anchor tables, held-out transfer-law errors, the trained/transfer split analysis, 1PL/2PL comparisons, and the item-level shift-versus-deletion likelihood tests.

A.3 Pre-registration

Timestamped commits of functional forms, claim hierarchy (T1/T2), discriminating experiments, and the falsifiable concentration prediction for 8B/14B/32B.